

Measurement-Oriented Evaluation of Multilayer Thin-Ferrite Cylindrical Shields for Low-Noise Magnetic Instrumentation

Author Name

Abstract—Low-noise magnetic instruments are limited not only by sensor noise, but also by shield-induced leakage fields and shield-related magnetic noise. This manuscript develops a measurement-oriented finite-element evaluation framework for multilayer thin-ferrite cylindrical shields intended for ultra-sensitive magnetic measurement systems. The shield is treated as a device under test in a calibrated field configuration rather than as a passive enclosure. The framework separates two finite-element problems: an external uniform-field model for directional shielding-factor evaluation and a virtual pickup-coil model for the geometry-dependent magnetic-noise-related loss indicator $\int H^2 dV$ (IntH2). A validated no-shield directional reference is used to compute SF_x , SF_z , and the corresponding leakage ratios from field components rather than from vector magnitude. A compact analytical scaling argument is included to connect the finite-element matrix to thin cylindrical-shell theory without replacing the numerical model. The method also defines layer-resolved IntH2 decomposition, exact ferrite-volume calculation, volume-normalized auxiliary metrics, and a reproducible data-export and consistency-check pipeline. The current implementation targets manufacturable multilayer cylindrical thin-ferrite shells with 100 mm inner radius, 200 mm length, and ferrite sheet thickness from 0.08 mm to 0.60 mm. Validation studies required for IEEE TIM submission, including mesh and boundary-domain convergence, axial-field checks, segmented-shell correction, permeability sensitivity, and optional prototype shielding-factor measurement, are specified as part of the evaluation protocol. Numerical conclusions are reported only for cases with real AEDT exports and post-processing validation.

Index Terms—Ferrite magnetic shield, finite-element method, instrumentation, magnetic noise, multilayer cylindrical shell, shielding factor.

I. INTRODUCTION

ULTRA-SENSITIVE magnetic measurement systems, including atomic magnetometers, spin-exchange-relaxation-free (SERF) comagnetometers, magnetoencephalography instruments, and precision tests of fundamental symmetries, require a controlled low-field and low-noise magnetic environment [1]–[5]. In such systems, the magnetic shield is part of the measurement chain. It determines the environmental-field leakage that reaches the sensor and can also generate magnetic-field noise through material loss. Therefore, the shield should not be evaluated only by a single shielding factor. For low-noise magnetic instrumentation, directional shielding factor, center

leakage field, ferrite volume, radial space occupation, and material-loss-related noise indicators should be evaluated together.

High-permeability metallic shields are widely used to attenuate static and low-frequency fields [6]–[8]. However, their high electrical conductivity produces eddy-current magnetic noise that can become relevant in the frequency band of high-sensitivity magnetometers [9]–[11]. Ferrites are attractive alternatives because their conductivity is much lower than that of metallic alloys. This suppresses eddy-current noise; the remaining low-frequency shield-induced noise is then associated mainly with magnetic loss, represented by the imaginary permeability μ_r'' , and with the spatial distribution of magnetic field inside the ferrite [9], [12]–[15]. A recent IEEE TIM study on cubic ferrite shields demonstrated the practical importance of ferrites in compact low-noise shielding [16]. Nevertheless, the cubic panel geometry does not directly address open-ended cylindrical shields, which are common around atomic sensors and compact comagnetometers.

A monolithic cylindrical ferrite shield is mechanically difficult to fabricate when the diameter is large and the wall is thin. Ferrites are brittle after sintering, and machining a 200-mm-diameter thin-walled cylinder with sub-millimeter wall thickness is costly and fragile. A manufacturable alternative is to assemble a shield from multilayer thin-ferrite cylindrical shells or segmented ferrite sheets. This introduces a coupled design problem: layer number N_L , ferrite thickness a , radial gap g , circumferential segmentation, ferrite volume V_f , and total radial occupation all affect the center leakage field and the geometry-dependent loss integral used in magnetic-noise calculations.

Existing studies provide important foundations, but do not fully cover this measurement-oriented design problem. Kornack *et al.* reported a low-noise ferrite cylinder with measured magnetic noise [12]. Lee and Romalis derived magnetic-noise formulas for high-permeability shields and conducting objects with simple geometry [9]. Later studies modeled cylindrical or multi-annular ferrite shields and discussed gap effects and material parameters [13], [14], [17]. However, previous work has not simultaneously reported a validated no-shield directional reference, layer-resolved IntH2 analysis, and a multilayer thin-ferrite cylindrical geometry under a unified evaluation framework. Table I summarizes the relationship between representative prior work and the present manuscript.

The novelty of this work is not simply the use of multiple ferrite layers. The contribution is a measurement-oriented

TABLE I
RELATIONSHIP TO REPRESENTATIVE FERRITE SHIELD STUDIES

Work	Geometry	Main quantity	Scope relative to this work
Kornack <i>et al.</i> [12]	Single ferrite cylinder	SF and measured noise	No multilayer thin-shell comparison
Lee and Romalis [9]	Simple analytic shapes	Magnetic-noise formula	No finite-length multilayer FEM geometry
Ma <i>et al.</i> [13]	Cylindrical ferrite shield	SF and IntH2	Layer-resolved IntH2 not reported
Lu <i>et al.</i> [14], [17]	Multi-annular/gapped ferrite shields	Noise-related FEM analysis	Directional no-shield reference not emphasized
Sun <i>et al.</i> [16]	Cubic ferrite shield	SF and measured noise	Cubic geometry, not cylindrical multilayer shell
This work	Multilayer thin-ferrite cylinder	Directional SF, V_f , layer-wise IntH2	Measurement-oriented FEM and data pipeline

evaluation framework that combines: 1) a validated no-shield directional reference for shielding-factor evaluation; 2) a virtual pickup-coil model for the geometry-dependent magnetic-noise-related loss integral; 3) layer-resolved IntH2 decomposition for multilayer shields; 4) fixed-volume comparison between single-layer and multilayer candidates; 5) unit-consistent auxiliary metrics for low-noise shield design; and 6) a reproducible raw-data, post-processing, validation, and plotting pipeline.

This draft focuses on the Introduction and Model/Method sections. Section II defines the coordinate system, two finite-element models, directional shielding factor, analytical scaling motivation, virtual pickup-coil loss integral, cylindrical geometry, simulation matrix, data-export pipeline, auxiliary metrics, validation studies, and optional experimental protocol.

II. MODEL AND EVALUATION METHODOLOGY

A. Coordinate System and Measurement Configuration

The shield is modeled as a device under test (DUT) placed at the center of a calibrated magnetic-field region. The cylinder axis is defined as the Cartesian z -axis, while x and y span the transverse plane. Because the shield is a finite open-ended cylinder, transverse and axial shielding are not equivalent. The quantities SF_x and SF_y describe transverse shielding through the sidewall, whereas SF_z describes axial shielding affected by leakage through the open ends [6], [27]–[29].

Two finite-element models are used, as shown schematically in Fig. 1. These models answer different measurement questions and should not be physically conflated. The external uniform-field model evaluates environmental-field attenuation and provides B_0 , B_{center} , SF_q , and LeakageRatio_q . The virtual pickup-coil model evaluates how material loss couples to the measured magnetic field through the fluctuation-dissipation theorem and provides $\text{IntH2}_{\text{total}}$ and IntH2_{L_i} .

B. Directional Shielding Factor

For an external field applied along $q \in \{x, y, z\}$, the no-shield reference field at the observation point is

$$\mathbf{B}_0 = (B_{0,x}, B_{0,y}, B_{0,z}), \quad (1)$$

and the shielded center leakage field is

$$\mathbf{B}_{\text{center}} = (B_{\text{center},x}, B_{\text{center},y}, B_{\text{center},z}). \quad (2)$$

Fig. 1 placeholder: measurement-oriented framework. External uniform-field model exports B_0 and B_{center} for SF calculation. Virtual pickup-coil model exports $\text{IntH2}_{\text{total}}$ and IntH2_{L_i} . Post-processing computes η_S^* , ρ_H , and χ_H .

Fig. 1. Measurement-oriented evaluation workflow. The final manuscript will replace this placeholder with a compact vector diagram.

The directional shielding factor is defined by

$$SF_q = \frac{|B_{0,q}|}{|B_{\text{center},q}|}, \quad (3)$$

and the corresponding leakage ratio is

$$\text{LeakageRatio}_q = \frac{|B_{\text{center},q}|}{|B_{0,q}|} = \frac{1}{SF_q}. \quad (4)$$

The formal shielding factor is computed from the intended directional component. The vector magnitude B_{Mag} is retained only as an auxiliary consistency check and is not used as the primary SF definition.

C. No-Shield Directional Reference

The no-shield model is the calibration reference for the external-field problem. The ferrite bodies are deleted or assigned vacuum while the air region, boundary dimensions, observation point, solver setting, and excitation amplitude are kept identical to the shielded model. In vacuum, for $H_0 = 1 \text{ A/m}$,

$$B_0 = \mu_0 H_0 = (4\pi \times 10^{-7})(1) = 1.256637 \times 10^{-6} \text{ T}. \quad (5)$$

Thus, an ideal x -directed reference should give $B_{0,x} \approx 1.256637 \times 10^{-6} \text{ T}$, with $B_{0,y}$ and $B_{0,z}$ much smaller. The residual transverse components are treated as numerical residuals caused by finite-element discretization, finite-domain boundary approximation, and interpolation at the observation point.

The reference field is accepted only if the intended component dominates the other components:

$$\Gamma_q = \frac{|B_{0,q}|}{\max_{p \neq q} |B_{0,p}|}. \quad (6)$$

In the processing pipeline, a failed directional reference prevents the corresponding shielding factors from being used in figures or conclusions.

D. Analytical Scaling Motivation

The finite-element model is the primary evaluation tool because the shield is finite in length, open ended, and may include layer gaps or segmented slots. Nevertheless, the choice of design variables is guided by the classical transverse cylindrical-shell solution. In a source-free region with linear isotropic permeability, $\nabla \times \mathbf{H} = 0$ and $\nabla \cdot \mathbf{B} = 0$ allow the magnetic scalar potential to satisfy Laplace's equation. For an infinitely long coaxial cylindrical shell in a transverse uniform field, the potential in each annular region can be written as

$$\phi_k(r, \theta) = \left(C_k r + \frac{D_k}{r} \right) \cos \theta, \quad (7)$$

where the constants are obtained from the continuity of normal B and tangential H at each interface [6]. For a single high-permeability thin shell, this solution gives the familiar leading transverse scaling

$$S_T \simeq 1 + \frac{\mu_r t}{2R}, \quad (8)$$

showing that shell thickness, mean radius, and permeability set the first-order shielding contribution.

For a multilayer coaxial shell, the analytical expansion contains direct layer terms and interlayer radius-coupling terms. A typical coupling factor between two layers with mean radii $R_i < R_j$ is

$$\Lambda_{ij} = 1 - \left(\frac{R_i}{R_j} \right)^2. \quad (9)$$

This motivates comparing layer number, radial gap, and total radial occupation rather than comparing total thickness alone. The present manuscript uses this analytical structure only as a scaling motivation; the reported SF_q values are obtained from the validated directional FEM reference in (3), not from the ideal infinite-cylinder expression.

Under the thin-shell approximation, a set of equal-height coaxial layers with total ferrite thickness $T_f = N_L a$ has

$$V_f \approx 2\pi L R_c T_f, \quad (10)$$

where R_c is the radial-window center radius. Therefore, fixed T_f is a useful first-order proxy for fixed ferrite volume. In the actual post-processing, however, V_f is always computed from the exact layer radii in (22) or from CAD volume export.

E. Virtual Pickup-Coil Model and IntH2

The shield-induced magnetic noise can be related to material loss through the fluctuation-dissipation theorem [9], [18]. A virtual pickup coil placed at the observation point produces a magnetic field H_{vc} in the ferrite. For coil area A , turn number N_c , peak current I_0 , and magnetic moment $m = AN_c I_0$, the hysteresis-loss contribution to the magnetic-field noise amplitude spectral density can be written as

$$\delta B_{\text{hyst}}(f) = \frac{1}{m} \left(\frac{2k_B T \mu''}{\pi f} \int_V H_{vc}^2 dV \right)^{1/2}, \quad (11)$$

where $\mu'' = \mu_0 \mu_r''$ is the absolute imaginary permeability if μ_r'' is specified as a relative quantity. The current I_0 is a peak current; if an RMS current is used in another implementation, the normalization must be converted consistently.

This manuscript does not claim measured magnetic-noise spectra. The exported FEM quantity is the geometry-dependent magnetic-noise-related loss indicator

$$\text{IntH2} = \int_V H_{vc}^2 dV, \quad (12)$$

with SI unit $A^2 m$. IntH2 becomes an absolute noise prediction only after measured $\mu_r''(f)$, temperature, frequency, and virtual-coil normalization are specified. Relative comparison by IntH2 is meaningful only for identical material properties, identical coil normalization, and identical observation direction.

For two candidate geometries A and B evaluated under identical material loss and virtual-coil normalization, (11) gives the relative loss-induced field-noise scaling

$$\frac{\delta B_A(f)}{\delta B_B(f)} = \left(\frac{\text{IntH2}_A}{\text{IntH2}_B} \right)^{1/2}. \quad (13)$$

Thus, IntH2 ratios should not be interpreted linearly as magnetic-noise ratios; the square-root relation applies only under the identical-normalization assumptions stated above.

For a multilayer shield, each ferrite layer is retained as an independent body. The layer-resolved terms are

$$\text{IntH2}_{L_i} = \int_{V_{L_i}} H_{vc}^2 dV, \quad (14)$$

and

$$\text{IntH2}_{\text{total}} = \sum_{i=1}^{N_L} \text{IntH2}_{L_i}. \quad (15)$$

The export is accepted only if

$$\frac{|\text{IntH2}_{\text{total}} - \sum_i \text{IntH2}_{L_i}|}{|\text{IntH2}_{\text{total}}|} < 10^{-3}. \quad (16)$$

F. Multilayer Cylindrical Geometry

The DUT is an open-ended multilayer cylindrical shell with inner radius R_{in} , axial length L , layer count N_L , ferrite thickness a , and radial interlayer gap g . The current design matrix uses

$$R_{\text{in}} = 100 \text{ mm}, \quad L = 200 \text{ mm}. \quad (17)$$

The total ferrite thickness and radial occupation are

$$T_f = N_L a, \quad (18)$$

$$T_{\text{space}} = N_L a + (N_L - 1)g. \quad (19)$$

For layer i , where $i = 1$ is the innermost layer,

$$R_{i,\text{in}} = R_{\text{in}} + (i - 1)(a + g), \quad (20)$$

$$R_{i,\text{out}} = R_{i,\text{in}} + a. \quad (21)$$

The ferrite volume is calculated from the exact cylindrical-shell expression

$$V_f = \pi L \sum_{i=1}^{N_L} (R_{i,\text{out}}^2 - R_{i,\text{in}}^2). \quad (22)$$

Fig. 2 placeholder: multilayer cylindrical shell geometry with R_{in} , L , a , g , N_L , T_f , and T_{space} .

Fig. 2. Geometry definition for the multilayer thin-ferrite cylindrical shell.

Although V_f is displayed in mm^3 for engineering readability, all SI-derived metrics are computed after unit conversion.

For segmented-shell correction, the circumferential coverage fraction is defined as

$$\eta_\phi = 1 - \frac{N_{\text{seg}} g_\phi}{2\pi R_{\text{mid}}}, \quad (23)$$

where N_{seg} is the number of circumferential segments, g_ϕ is the slot width, and R_{mid} is the layer mid-radius. Aligned and staggered slot patterns are evaluated separately when the segmented AEDT models are available.

G. Simulation Matrix and Data Export

The present main matrix is a transverse continuous-shell screening study. It is intentionally separated from axial, segmented, permeability-sensitivity, and convergence studies so that each validation stage can be tracked independently. Table II lists the current main cases.

Each shielded external-field case exports $B_{\text{center},x}$, $B_{\text{center},y}$, $B_{\text{center},z}$, and $B_{\text{center},\text{Mag}}$. The no-shield reference exports $B_{0,x}$, $B_{0,y}$, $B_{0,z}$, and $B_{0,\text{Mag}}$. Each virtual pickup-coil case exports $\text{IntH2}_{\text{total}}$ and the layerwise quantities IntH2_{L_i} . The post-processing scripts merge raw CSV exports by case ID and field direction, compute the derived metrics, and mark missing or failed rows without substituting zero values.

H. Comparison Metrics

The volume-normalized shielding index is defined as

$$\eta_S = \frac{\ln(\text{SF}_x)}{V_f}. \quad (24)$$

It is an auxiliary index introduced for this comparison, not a material property. When V_f is reported in mm^3 , the displayed unit is mm^{-3} ; SI-derived quantities are computed after converting volume to m^3 .

A dimensionless normalized form is

$$\eta_S^* = \frac{\ln(\text{SF}_x)/V_f}{\ln(\text{SF}_{x,\text{ref}})/V_{\text{ref}}}, \quad (25)$$

where the reference is the continuous single-layer shell with $a = 0.60 \text{ mm}$. The loss-integral density and attenuation-normalized loss indicator are

$$\rho_H = \frac{\text{IntH2}_{\text{total}}}{V_f[\text{m}^3]}, \quad (26)$$

$$\chi_H = \frac{\text{IntH2}_{\text{total}}}{\ln(\text{SF}_x)}. \quad (27)$$

The metrics are interpreted jointly with T_{space} and the validation status. No single auxiliary metric defines a final shield geometry.

I. Required Validation Studies

Before the manuscript is extended to final results, the following validation studies are required.

Mesh convergence: at least C1_N1_t008_x and C2_N4_t015_g008_x are solved with coarse, medium, and fine mesh settings. The processed table records total element count, elements across ferrite thickness, $B_{\text{center},x}$, SF_x , $\text{IntH2}_{\text{total}}$, and relative changes. The target criteria are $\Delta \text{SF}_x < 1\%$, $\Delta \text{IntH2}_{\text{total}} < 2\%$, and $\Delta B_{\text{center},x} < 1\%$ from medium to fine mesh.

Boundary-domain convergence: C2_N4_t015_g008_x is evaluated with air-domain scales of $2R$, $3R$, and $5R$; optional $7R$ is retained as an additional check. The target criteria are less than 1% change in SF_x and less than 2% change in $\text{IntH2}_{\text{total}}$ from $3R$ to $5R$.

Axial shielding: a validated B0_reference_z model is required before shielded SF_z is computed. The selected equal-volume cases are C1_N1_t060_z, C2_N3_t020_g010_z, and C2_N4_t015_g008_z. The transverse ranking is not generalized to axial shielding unless these cases support it.

Segmented-shell correction: C2_N4_t015_g008_x is compared with segmented aligned and staggered slot models using $N_{\text{seg}} = 12$ and $g_\phi = 1.6 \text{ mm}$. The segmented-to-continuous penalty is quantified by $\text{SF}_{x,\text{seg}}/\text{SF}_{x,\text{cont}}$ and $\text{IntH2}_{\text{seg}}/\text{IntH2}_{\text{cont}}$.

Permeability sensitivity: C1_N1_t060_x and C2_N4_t015_g008_x are evaluated at $\mu'_r = 500, 1000, 2000$, and 5000 . The screening value $\mu'_r = 1000$ is not treated as a measured material constant.

Field-map export: the final paper should include external-field B distributions, flux lines, H_{vc} contours, and H_{vc}^2 contours to support the interpretation of flux redistribution and layerwise IntH2 contributions.

Prototype SF protocol: if hardware is available, the prototype SF measurement uses a calibrated Helmholtz or triaxial coil system and a calibrated magnetic sensor. For independent B_0 and B_{center} measurements,

$$\frac{u(\text{SF}_{q,\text{exp}})}{\text{SF}_{q,\text{exp}}} = \sqrt{\left[\frac{u(B_{0,q})}{B_{0,q}} \right]^2 + \left[\frac{u(B_{\text{center},q})}{B_{\text{center},q}} \right]^2}. \quad (28)$$

For correlated measurements,

$$u_r^2(\text{SF}_{q,\text{exp}}) = u_r^2(B_0) + u_r^2(B_{\text{center}}) - 2\rho u_r(B_0)u_r(B_{\text{center}}). \quad (29)$$

If no experimental data are available, this section remains a protocol and is not presented as completed validation.

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TABLE II
MAIN CONTINUOUS-SHELL TRANSVERSE MATRIX

Case	N_L	a (mm)	g (mm)	Direction	Purpose
B0_reference_x	0	0	0	x	no-shield reference
C1_N1_t008_x	1	0.08	0	x	thinnest single layer
C1_N1_t020_x	1	0.20	0	x	thin single layer
C1_N1_t040_x	1	0.40	0	x	intermediate single layer
C1_N1_t060_x	1	0.60	0	x	equal-volume single-layer baseline
C2_N2_t020_g010_x	2	0.20	0.10	x	multilayer candidate
C2_N3_t020_g010_x	3	0.20	0.10	x	equal-volume multilayer candidate
C2_N4_t015_g008_x	4	0.15	0.08	x	equal-volume multilayer candidate

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